容斥原理

定理（Inclusion-exclusion principle，容斥原理，并集版本）$A_1,\cdots,A_n$是任意的有限集合，则有：

$$\left|\bigcup_{t=1}^{n} A_t\right| = \sum_{k=1}^{n}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le n}\left|A_{i_1}\cap\cdots\cap A_{i_k}\right|\right)$$

Proof：当$n=1$时，左边$=|A_1|$，右边$=(-1)^2\sum_{1\le i_1\le 1}|A_{i_1}|=|A_1|$

左边=右边　∴$n=1$时定理得证.

当$n=2$时，左边$=\left|\bigcup_{t=1}^{2}A_t\right|=|A_1\cup A_2|$

$$右边=\sum_{k=1}^{2}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le 2}|A_{i_1}\cap\cdots\cap A_{i_k}|\right)=(-1)^2\sum_{1\le i_1\le 2}|A_{i_1}|+(-1)^3\sum_{1\le i_1<i_2\le 2}|A_{i_1}\cap A_{i_2}|$$

$$=|A_1|+|A_2|-|A_1\cap A_2|$$

这种情况我们满足于画 Venn diagram 的证明.　∴左边=右边，$n=2$时定理得证.

当$n=3$时，左边$=\left|\bigcup_{t=1}^{3}A_t\right|=|A_1\cup A_2\cup A_3|$

$$右边=\sum_{k=1}^{3}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le 3}|A_{i_1}\cap\cdots\cap A_{i_k}|\right)$$

$$=(-1)^2\sum_{1\le i_1\le 3}|A_{i_1}|+(-1)^3\sum_{1\le i_1<i_2\le 3}|A_{i_1}\cap A_{i_2}|+(-1)^4\sum_{1\le i_1<i_2<i_3\le 3}|A_{i_1}\cap A_{i_2}\cap A_{i_3}|$$

$$=|A_1|+|A_2|+|A_3|-|A_1\cap A_2|-|A_1\cap A_3|-|A_2\cap A_3|+|A_1\cap A_2\cap A_3|$$

$$左边=|A_1\cup A_2\cup A_3|=|(A_1\cup A_2)\cup A_3|=|A_1\cup A_2|+|A_3|-|(A_1\cup A_2)\cap A_3|$$

$$=|A_1|+|A_2|-|A_1\cap A_2|+|A_3|-|(A_1\cap A_3)\cup(A_2\cap A_3)|$$

$$=|A_1|+|A_2|-|A_1\cap A_2|+|A_3|-\left(|A_1\cap A_3|+|A_2\cap A_3|-|A_1\cap A_2\cap A_3|\right)$$

$= |A_1| + |A_2| - |A_1 \cap A_2| + |A_3| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3|$

$= |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3|$

$=$ 右边　　$\therefore$ 左边 $=$ 右边，$n=3$ 时定理得证.

假设当 $n=\lambda$（$\lambda \in \mathbb{Z}$，$\lambda \geq 3$）时定理得证，即有：

$$\left|\bigcup_{t=1}^{\lambda} A_t\right| = \sum_{k=1}^{\lambda}(-1)^{k+1}\left(\sum_{1 \leq i_1 < \cdots < i_k \leq \lambda} |A_{i_1} \cap \cdots \cap A_{i_k}|\right)$$

则当 $n=\lambda+1$ 时，左边 $= \left|\bigcup_{t=1}^{\lambda+1} A_t\right|$

右边 $= \sum_{k=1}^{\lambda+1}(-1)^{k+1}\left(\sum_{1 \leq i_1 < \cdots < i_k \leq \lambda+1} |A_{i_1} \cap \cdots \cap A_{i_k}|\right)$

$\therefore$ 左边 $= \left|\bigcup_{t=1}^{\lambda+1} A_t\right| = |A_1 \cup \cdots \cup A_{\lambda+1}| = |(A_1 \cup \cdots \cup A_\lambda) \cup A_{\lambda+1}|$

$$= \left|\left(\bigcup_{t=1}^{\lambda} A_t\right) \cup A_{\lambda+1}\right| = \left|\bigcup_{t=1}^{\lambda} A_t\right| + |A_{\lambda+1}| - \left|\left(\bigcup_{t=1}^{\lambda} A_t\right) \cap A_{\lambda+1}\right|$$

$$= \sum_{k=1}^{\lambda}(-1)^{k+1}\left(\sum_{1 \leq i_1 < \cdots < i_k \leq \lambda} |A_{i_1} \cap \cdots \cap A_{i_k}|\right) + |A_{\lambda+1}| - \left|\bigcup_{t=1}^{\lambda} (A_t \cap A_{\lambda+1})\right|$$

$$= \sum_{k=1}^{\lambda}(-1)^{k+1}\left(\sum_{1 \leq i_1 < \cdots < i_k \leq \lambda} |A_{i_1} \cap \cdots \cap A_{i_k}|\right) + |A_{\lambda+1}|$$

$$- \sum_{k=1}^{\lambda}(-1)^{k+1}\left(\sum_{1 \leq i_1 < \cdots < i_k \leq \lambda} |(A_{i_1} \cap A_{\lambda+1}) \cap \cdots \cap (A_{i_k} \cap A_{\lambda+1})|\right)$$

$$= \sum_{k=1}^{\lambda}(-1)^{k+1}\left(\sum_{1 \leq i_1 < \cdots < i_k \leq \lambda} |A_{i_1} \cap \cdots \cap A_{i_k}|\right) + |A_{\lambda+1}| + \sum_{k=1}^{\lambda}(-1)^{k+2}\left(\sum_{1 \leq i_1 < \cdots < i_k \leq \lambda} |A_{i_1} \cap \cdots \cap A_{i_k} \cap A_{\lambda+1}|\right)$$

$$=(-1)^2\sum_{1\le i_1\le\lambda}|A_{i_1}|+|A_{\lambda+1}|$$

$$+(-1)^3\sum_{1\le i_1<i_2\le\lambda}|A_{i_1}\cap A_{i_2}|+(-1)^3\sum_{1\le i_1\le\lambda}|A_{i_1}\cap A_{\lambda+1}|$$

$$+(-1)^4\sum_{1\le i_1<i_2<i_3\le\lambda}|A_{i_1}\cap A_{i_2}\cap A_{i_3}|+(-1)^4\sum_{1\le i_1<i_2\le\lambda}|A_{i_1}\cap A_{i_2}\cap A_{\lambda+1}|$$

$$+(-1)^5\sum_{1\le i_1<i_2<i_3<i_4\le\lambda}|A_{i_1}\cap A_{i_2}\cap A_{i_3}\cap A_{i_4}|+(-1)^5\sum_{1\le i_1<i_2<i_3\le\lambda}|A_{i_1}\cap A_{i_2}\cap A_{i_3}\cap A_{\lambda+1}|$$

$$+\cdots$$

$$+(-1)^{\lambda+1}\sum_{1\le i_1<\cdots<i_\lambda\le\lambda}|A_{i_1}\cap\cdots\cap A_{i_\lambda}|+(-1)^{\lambda+1}\sum_{1\le i_1<\cdots<i_{\lambda-1}\le\lambda}|A_{i_1}\cap\cdots\cap A_{i_{\lambda-1}}\cap A_{\lambda+1}|$$

$$+(-1)^{\lambda+2}\sum_{1\le i_1<\cdots<i_\lambda\le\lambda}|A_{i_1}\cap\cdots\cap A_{i_\lambda}\cap A_{\lambda+1}|$$

$$=(-1)^2\sum_{1\le i_1\le\lambda+1}|A_{i_1}|+(-1)^3\sum_{1\le i_1<i_2\le\lambda+1}|A_{i_1}\cap A_{i_2}|$$

$$+(-1)^4\sum_{1\le i_1<i_2<i_3\le\lambda+1}|A_{i_1}\cap A_{i_2}\cap A_{i_3}|+(-1)^5\sum_{1\le i_1<i_2<i_3<i_4\le\lambda+1}|A_{i_1}\cap A_{i_2}\cap A_{i_3}\cap A_{i_4}|$$

$$+\cdots$$

$$+(-1)^{\lambda+1}\sum_{1\le i_1<\cdots<i_\lambda\le\lambda+1}|A_{i_1}\cap\cdots\cap A_{i_\lambda}|+(-1)^{\lambda+2}\sum_{1\le i_1<\cdots<i_{\lambda+1}\le\lambda+1}|A_{i_1}\cap\cdots\cap A_{i_{\lambda+1}}|$$

$$=\sum_{k=1}^{\lambda+1}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le\lambda+1}|A_{i_1}\cap\cdots\cap A_{i_k}|\right)=\text{右边}.$$

$\therefore$ 对 $\forall n\in\mathbb{Z}_{\geq 1}$，定理得证 $\square$

定理（容斥原理的交集形式）$A_1, \cdots, A_n$ 是任意的有限集合，则有：

$$\left|\bigcap_{t=1}^{n} A_t\right| = \sum_{k=1}^{n}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le n}\left|A_{i_1}\cup\cdots\cup A_{i_k}\right|\right)$$

Proof：当 $n=1$ 时，左边 $=|A_1|$，右边 $=(-1)^2\sum_{1\le i_1\le 1}|A_{i_1}| = |A_1|$ $\therefore$ 左边 $=$ 右边.

当 $n=2$ 时，左边 $=|A_1\cap A_2|$，右边 $=\sum_{k=1}^{2}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le 2}\left|A_{i_1}\cup\cdots\cup A_{i_k}\right|\right)$

$= |A_1| + |A_2| - |A_1\cup A_2|$

我们满足于画 Venn diagram 来证明. $\therefore$ 左边 $=$ 右边.

假设当 $n=\lambda$ ($\lambda\in\mathbb{Z}$，$\lambda\ge 2$) 时定理得证，即有：

$$\left|\bigcap_{t=1}^{\lambda} A_t\right| = \sum_{k=1}^{\lambda}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le \lambda}\left|A_{i_1}\cup\cdots\cup A_{i_k}\right|\right)$$

则当 $n=\lambda+1$ 时，左边 $=\left|\bigcap_{t=1}^{\lambda+1} A_t\right|$，右边 $=\sum_{k=1}^{\lambda+1}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le \lambda+1}\left|A_{i_1}\cup\cdots\cup A_{i_k}\right|\right)$

$\because$ 左边 $=\left|\bigcap_{t=1}^{\lambda+1} A_t\right| = \left|\left(\bigcap_{t=1}^{\lambda} A_t\right)\cap A_{\lambda+1}\right| = \left|\bigcap_{t=1}^{\lambda} A_t\right| + |A_{\lambda+1}| - \left|\left(\bigcap_{t=1}^{\lambda} A_t\right)\cup A_{\lambda+1}\right|$

$$= \sum_{k=1}^{\lambda}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le \lambda}\left|A_{i_1}\cup\cdots\cup A_{i_k}\right|\right) + |A_{\lambda+1}| - \left|\bigcap_{t=1}^{\lambda}(A_t\cup A_{\lambda+1})\right|$$

$$= \sum_{k=1}^{\lambda}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le \lambda}\left|A_{i_1}\cup\cdots\cup A_{i_k}\right|\right) + |A_{\lambda+1}|$$

$$- \sum_{k=1}^{\lambda}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le \lambda}\left|(A_{i_1}\cup A_{\lambda+1})\cup\cdots\cup(A_{i_k}\cup A_{\lambda+1})\right|\right)$$

$$= \sum_{k=1}^{\lambda}(-1)^{k+1}\left(\sum_{1\le i_1<\cdots<i_k\le \lambda}\left|A_{i_1}\cup\cdots\cup A_{i_k}\right|\right) + |A_{\lambda+1}| + \sum_{k=1}^{\lambda}(-1)^{k+2}\left(\sum_{1\le i_1<\cdots<i_k\le \lambda}\left|A_{i_1}\cup\cdots\cup A_{i_k}\cup A_{\lambda+1}\right|\right)$$

$$= (-1)^2 \sum_{1 \le i_1 \le \lambda+1} |A_{i_1}|$$

$$+ (-1)^3 \sum_{1 \le i_1 < i_2 \le \lambda} |A_{i_1} \cup A_{i_2}| + (-1)^3 \sum_{1 \le i_1 \le \lambda} |A_{i_1} \cup A_{\lambda+1}|$$

$$+ (-1)^4 \sum_{1 \le i_1 < i_2 < i_3 \le \lambda} |A_{i_1} \cup A_{i_2} \cup A_{i_3}| + (-1)^4 \sum_{1 \le i_1 < i_2 \le \lambda} |A_{i_1} \cup A_{i_2} \cup A_{\lambda+1}|$$

$$+ \cdots$$

$$+ (-1)^{\lambda+1} \sum_{1 \le i_1 < \cdots < i_\lambda \le \lambda} |A_{i_1} \cup \cdots \cup A_{i_\lambda}| + (-1)^{\lambda+1} \sum_{1 \le i_1 < \cdots < i_{\lambda-1} \le \lambda} |A_{i_1} \cup \cdots \cup A_{i_{\lambda-1}} \cup A_{\lambda+1}|$$

$$+ (-1)^{\lambda+2} \sum_{1 \le i_1 < \cdots < i_\lambda \le \lambda} |A_{i_1} \cup \cdots \cup A_{i_\lambda} \cup A_{\lambda+1}|$$

$$= (-1)^2 \sum_{1 \le i_1 \le \lambda+1} |A_{i_1}| + (-1)^3 \sum_{1 \le i_1 < i_2 \le \lambda+1} |A_{i_1} \cup A_{i_2}|$$

$$+ (-1)^4 \sum_{1 \le i_1 < i_2 < i_3 \le \lambda+1} |A_{i_1} \cup A_{i_2} \cup A_{i_3}| + \cdots$$

$$+ (-1)^{\lambda+1} \sum_{1 \le i_1 < \cdots < i_\lambda \le \lambda+1} |A_{i_1} \cup \cdots \cup A_{i_\lambda}| + (-1)^{\lambda+2} \sum_{1 \le i_1 < \cdots < i_{\lambda+1} \le \lambda+1} |A_{i_1} \cup \cdots \cup A_{i_{\lambda+1}}|$$

$=$ 右边.

$\therefore$ 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 定理得证. $\square$